RATIOS AND COMMON-SIZE ANALYSIS

Learning Outcome

- calculate and interpret common-size balance sheets and related financial ratios

Analysis of a company's balance sheet can provide insight into the company's liquidity and solvency—as of the balance sheet date—as well as the economic resources the company controls. **Liquidity** refers to a company's ability to meet its short-term financial commitments. Assessments of liquidity focus on a company's ability to convert assets to cash to pay for operating needs. **Solvency** refers to a company's ability to meet its financial obligations over the longer term. Assessments of solvency focus on the company's financial structure and its ability to pay long-term financing obligations. This lesson describes two tools for analyzing the balance sheet: common-size analysis and balance sheet ratios.

Common-Size Analysis of the Balance Sheet

The first technique, vertical common-size analysis, involves stating each balance sheet item as a percentage of total assets.[9] Common-size balance sheets are useful in comparing a company's balance sheet composition over time (time-series analysis) and across companies in the same industry. To illustrate, Panel A of Exhibit 10 presents balance sheets for three hypothetical companies. Company C, with assets of USD9.75 million is much larger than Company A and Company B, each with only USD3.25 million in assets. The common-size balance sheet presented in Panel B facilitates a comparison of these different-size companies.

Exhibit 10:

Balance Sheets for Companies A, B, and C

Panel A: Balance Sheets			
(in thousands of US dollars)	**A**	**B**	**C**
ASSETS			
Current assets			
Cash and cash equivalents	1,000	200	3,000
Short-term marketable securities	900	—	300
Accounts receivable	500	1,050	1,500
Inventory	100	950	300
Total current assets	2,500	2,200	5,100
Property, plant, and equipment, net	750	750	4,650
Intangible assets	—	200	—
Goodwill	—	100	—
Total assets	3,250	3,250	9,750
LIABILITIES AND SHAREHOLDERS' EQUITY			
Current liabilities			
Accounts payable	—	2,500	600
Total current liabilities	—	2,500	600
Long-term bonds payable	10	10	9,000
Total liabilities	10	2,510	9,600
Total shareholders' equity	3,240	740	150
Total liabilities and shareholders' equity	3,250	3,250	9,750

ASSETS			
Current assets			
Cash and cash equivalents	30.8	6.2	30.8
Short-term marketable securities	27.7	0.0	3.1
Accounts receivable	15.4	32.3	15.4
Inventory	3.1	29.2	3.1
Total current assets	76.9	67.7	52.3
Property, plant, and equipment, net	23.1	23.1	47.7
Intangible assets	0.0	6.2	0.0
Goodwill	0.0	3.1	0.0
Total assets	100.0	100.0	100.0
LIABILITIES AND SHAREHOLDERS' EQUITY			
Current liabilities			
Accounts payable	0.0	76.9	6.2
Total current liabilities	0.0	76.9	6.2
Long-term bonds payable	0.3	0.3	92.3
Total liabilities	0.3	77.2	98.5
Total shareholders' equity	99.7	22.8	1.5
Total liabilities and shareholders' equity	100.0	100.0	100.0

Most of the assets of Company A and B are current assets; however, Company A has nearly 60 percent of its total assets in cash and short-term marketable securities, whereas Company B has only 6 percent of its assets in cash. Company A is more liquid than Company B. Company A shows no current liabilities (its current liabilities round to less than USD10,000), and it has cash on hand of USD1.0 million to meet any near-term financial obligations it might have. In contrast, Company B has USD2.5 million of current liabilities, which exceed its available cash of only USD200,000. To pay those near-term obligations, Company B will need to collect some of its accounts receivables, sell more inventory, borrow from a bank, or raise more long-term capital (e.g., by issuing more bonds or more equity). Company C also appears more liquid than Company B. It holds more than 30 percent of its total assets in cash and short-term marketable securities, and its current liabilities are only 6.2 percent of the amount of total assets.

Company C's USD3.3 million in cash and short-term marketable securities is substantially more than its current liabilities of USD600,000. Turning to the question of solvency, however, note that 98.5 percent of Company C's assets are financed with liabilities. If Company C experiences significant fluctuations in cash flows, it may be unable to pay the interest and principal on its long-term bonds. Company A is far more solvent than Company C, with less than 1 percent of its assets financed with liabilities.

These examples are hypothetical only. Other than general comparisons, little more can be said without further detail. In practice, a wide range of factors affect a company's liquidity management and capital structure. The study **capital structure** is a fundamental issue addressed in Corporate Issuers modules.

Common-size balance sheets can also highlight differences in companies' strategies. Comparing the asset composition of the companies, Company C has made a greater proportional investment in property, plant, and equipment (PP&E)—possibly because it manufactures more of its products in-house. The presence of goodwill on Company B's balance sheet signifies that it has made one or more acquisitions in the past. In contrast, the lack of goodwill on the balance sheets of Company A and Company C suggests that these two companies may have pursued a strategy of internal growth rather than growth by acquisition. Company A may be in either a start-up or liquidation stage of operations as evidenced by the composition of its balance sheet. It has relatively little inventory and no accounts payable. It either has not yet established trade credit or it is in the process of paying off its obligations in the process of liquidating.

1. Based on the information presented in Exhibits 2, 6, and 8, which of the following items increased as a percentage of total assets from 2016 to 2017? (Note: More than one answer may be correct.)

   A. Total current assets
   B. Total financial liabilities
   C. Cash and cash equivalents

Hide Solution

Solution:

A and C are correct.

Total current assets increased from 26.1 percent of total assets in 2016 (EUR11,564 ÷ EUR44,277) to 28.1 percent in 2017 (EUR11,930 ÷ EUR42,497).

Cash and cash equivalents increased from 8.4 percent of total assets in 2016 (EUR3,702 EUR44,277) to 9.4 percent in 2017 (EUR4,011 ÷ EUR42,497).

Total financial liabilities decreased in 2017 both in absolute euro amounts (EUR5,034) and as a percentage of total assets (EUR5,034 ÷ EUR42,497 = 11.8%) when compared with 2016 (EUR6,481 ÷ EUR44,277 = 14.6%).

Overall, aspects of the company's liquidity position are somewhat stronger in 2017 than in 2016. The company's cash balances as a percentage of total assets increased. While current liabilities increased as a percentage of total assets and total liabilities remained approximately the same percentage, the mix of liabilities shifted. Financial liabilities, which represent future cash outlays, decreased as a percentage of total assets.

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Common-size analysis of the balance sheet is particularly useful in cross-sectional analysis—comparing companies to each other for a particular time period or comparing a company with industry or sector data. The analyst could select individual peer companies for comparison, use industry data from published sources, or compile data from databases. When analyzing a company, many analysts prefer to select the peer companies for comparison or to compile their own industry statistics.

Exhibit 11 presents common-size balance sheet data compiled for the 10 sectors of the S&P 500 using 2017 data. The sector classification follows the S&P/MSCI Global Industrial Classification System (GICS). The exhibit presents mean and median common-size balance sheet data for those companies in the S&P 500 for which 2017 data was available in the Compustat database.[10]

Exhibit 11:

Common-Size Balance Sheet Statistics for the S&P 500 Grouped by S&P/MSCI GICS Sector, 2017 (in percent except for No. of Observations)

					Panel A. Median Data		
	10	15	20	25	30	35	40
	Energy	Materials	Industrials	Consumer Discretionary	Consumer Staples	Health Care	Financia
Number of observation	34	27	68	81	33	59	64

	Energy	Materials	Industrials	Discretionary	Staples	Care	Financia
Cash and short-term investments	6.8%	6.3%	8.1%	8.3%	4.1%	11.2%	6.2%
Receivables	5.8%	8.8%	12.9%	6.8%	6.5%	9.7%	20.4%
Inventories	1.6%	8.9%	6.9%	14.9%	9.6%	4.3%	0.0%
Total current assets	16.1%	26.0%	30.5%	41.5%	29.1%	31.4%	N.A.
PP&E	73.3%	36.3%	12.5%	19.8%	17.2%	8.1%	0.9%
Intangibles	1.6%	27.9%	33.3%	16.8%	41.9%	37.6%	2.8%
Goodwill	*0.7%*	*20.0%*	*28.3%*	*11.3%*	*26.2%*	*22.8%*	*2.2%*
Accounts payable	5.7%	7.3%	6.2%	8.0%	8.0%	3.1%	27.0%
Current liabilities	10.9%	16.5%	22.5%	25.8%	25.0%	16.5%	N.A.
Long-term debt	27.3%	31.4%	28.0%	28.7%	32.3%	24.3%	6.4%
Total liabilities	49.3%	64.2%	65.5%	64.9%	63.8%	59.2%	86.7%
Common equity	47.3%	33.8%	34.5%	34.7%	36.2%	39.4%	12.6%
Preferred stock	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total equity	47.3%	33.8%	34.5%	34.7%	36.2%	39.4%	13.2%

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(Continued from previous Exhibit)

				Panel B. Mean Data (Continued from Previ			
	10	15	20	25	30	35	40
	Energy	Materials	Industrials	Consumer Discretionary	Consumer Staples	Health Care	Finan
Number of observations	34	27	68	81	33	59	64
Cash and short-term investments	6.9%	7.4%	9.2%	12.9%	7.3%	15.4%	11.2
Receivables	6.6%	10.5%	15.2%	9.0%	7.7%	11.2%	31.5
Inventories	3.4%	9.3%	7.8%	18.3%	10.6%	6.3%	3.8
Total current assets	17.7%	28.8%	32.9%	40.6%	27.8%	36.4%	N.A
PP&E	68.0%	36.9%	24.5%	25.1%	21.6%	11.2%	2.1
Intangibles	7.8%	26.6%	35.6%	23.0%	43.6%	43.9%	11.4
Goodwill	5.4%	18.4%	26.8%	14.6%	24.6%	27.3%	7.7
Accounts payable	5.9%	8.1%	7.1%	11.8%	9.8%	8.1%	35.9
Current liabilities	11.8%	17.0%	23.0%	26.8%	24.6%	21.2%	N.A
Long-term debt	28.3%	31.2%	29.4%	31.3%	32.4%	28.5%	10.3
Total liabilities	50.3%	63.4%	67.1%	67.5%	68.3%	60.1%	80.1
Common equity	46.4%	34.2%	32.3%	32.3%	30.9%	38.9%	18.2
Preferred stock	0.0%	0.0%	0.1%	0.0%	0.0%	0.1%	0.4
Total equity	46.4%	34.2%	32.4%	32.3%	30.9%	39.0%	18.5

PPE = Property, plant, and equipment, LT = Long term.
Source: Based on data from Compustat.

some preferred stock.

- Financial companies have the greatest percentage of total liabilities. Financial companies typically have relatively high financial leverage.
- Utility and real estate companies have the lowest level of receivables.
- Inventory levels are highest for consumer discretionary. Materials and consumer staples have the next highest inventories.
- Information technology companies use the least amount of leverage as evidenced by the lowest percentages for long-term debt and total liabilities and highest percentages for common and total equity.

Example 1 discusses an analyst using cross-sectional common-size balance sheet data.

EXAMPLE 1

Cross-Sectional Common-Size Analysis

Jason Lu is comparing two companies in the computer industry to evaluate their relative financial position as reflected on their balance sheets. He has compiled the following vertical common-size data for Apple and Microsoft, which is presented in Exhibit 12.

Exhibit 12:

Cross-Sectional Analysis: Consolidated Balance Sheets (as percent of total assets)

	Apple	Microsoft
ASSETS:	**30 September 2017**	**30 June 2017**
Current assets:		
Cash and cash equivalents	5.4	3.2
Short-term marketable securities	14.4	52.0
Accounts receivable	4.8	8.2
Inventories	1.3	0.9
Vendor non-trade receivables	4.7	0.0
Other current assets	3.7	2.0
Total current assets	34.3	66.3
Long-term marketable securities	51.9	2.5
Property, plant, and equipment, net	9.0	9.8
Goodwill	1.5	14.6
Acquired intangible assets, net	0.6	4.2
Other assets	2.7	2.6
Total assets	100.0	100.0
LIABILITIES AND SHAREHOLDERS' EQUITY:		
Current liabilities:		
Accounts payable	13.1	3.1
Short-term debt	3.2	3.8
Current portion of long-term debt	1.7	0.4
Accrued expenses	6.9	2.7
Deferred revenue	2.0	14.1
Other current liabilities	0.0	2.6
Total current liabilities	26.9	26.8
Long-term debt	25.9	31.6
Deferred revenue non-current	0.8	4.3

Total liabilities	64.3	70.0
Commitments and contingencies		
Total shareholders' equity	35.7	30.0
Total liabilities and shareholders' equity	100.0	100.0

Source: Based on data from companies' annual reports.

From these data, Lu learns the following:

- Apple and Microsoft have high levels of cash and short-term marketable securities, consistent with the information technology sector as reported in Exhibit 11. Apple also has a high balance in long-term marketable securities. This may reflect the success of the company's business model, which has generated large operating cash flows in recent years.
- Apple's level of accounts receivable is lower than Microsoft's and lower than the industry average. Further research is necessary to learn the extent to which this is related to Apple's cash sales through its own retail stores. An alternative explanation would be that the company has been selling or factoring receivables to a greater degree than the other companies; however, that explanation is unlikely given Apple's cash position. Additionally, Apple shows vendor non-trade receivables, reflecting arrangements with its contract manufacturers.
- Apple and Microsoft both have low levels of inventory, similar to industry medians as reported in Exhibit 11. Apple uses contract manufacturers and can rely on suppliers to hold inventory until needed. Additionally, in the Management Discussion and Analysis section of their annual report, Apple discloses USD38 billion of noncancelable manufacturing purchase obligations, USD33 billion of which is due within 12 months. These amounts are not currently recorded as inventory and reflect the use of contract manufacturers to assemble and test some finished products. The use of purchase commitments and contract manufacturers implies that inventory may be "understated." Microsoft's low level of inventory is consistent with its business mix, which is more heavily weighted to software than to hardware.
- Apple and Microsoft have a level of PP&E that is relatively close to the sector median, as reported in Exhibit 11.
- Apple has a very low amount of goodwill, reflecting its strategy to grow organically rather than through acquisition. Microsoft's level of goodwill, while higher than Apple's, is lower than the industry median and mean. Microsoft made a number of major acquisitions (e.g., Nokia in 2014), but subsequently (in 2015) it wrote off significant amounts of goodwill as an impairment charge.
- Apple's level of accounts payable is higher than the computer industry average, but given the company's high level of cash and investments, it is unlikely that this is a problem.
- Apple's and Microsoft's levels of long-term debt are slightly higher than industry averages. Again, given the companies' high level of cash and investments, it is unlikely that this is a problem.

Balance Sheet Ratios

Ratios facilitate time-series and cross-sectional analysis of a company's financial position. **Balance sheet ratios** are those involving balance sheet items only. Each of the line items on a vertical common-size balance sheet is a ratio in that it expresses a balance sheet amount in relation to total assets. Other balance sheet ratios compare one balance sheet item to another. For example, the current ratio expresses current assets in relation to current liabilities as an indicator of a company's liquidity. Balance sheet ratios include **liquidity ratios** (measuring the company's ability to meet its short-term obligations) and **solvency ratios** (measuring the company's ability to meet long-term and other

Exhibit 13:

Balance Sheet Ratios

Liquidity Ratios	Calculation	Indicates
Current	Current assets ÷ Current liabilities	Ability to meet current liabilities
Quick (acid test)	(Cash + Marketable securities + Receivables) ÷ Current liabilities	Ability to meet current liabilities
Cash	(Cash + Marketable securities) ÷ Current liabilities	Ability to meet current liabilities
Solvency Ratios		
Long-term debt-to-equity	Total long-term debt ÷ Total equity	Financial risk and financial leverage
Debt-to-equity	Total debt ÷ Total equity	Financial risk and financial leverage
Total debt	Total debt ÷ Total assets	Financial risk and financial leverage
Financial leverage	Total assets ÷ Total equity	Financial risk and financial leverage

Ratio Analysis

1. Based on its balance sheet presented earlier, the current ratio for SAP Group at 31 December 2017 is *closest* to:

   A. 1.17.
   B. 1.20.
   C. 2.00.

   Hide Solution

   Solution:

   A is correct. SAP Group's current ratio (Current assets ÷ Current liabilities) at 31 December 2017 is 1.17 (EUR11,930 million ÷ EUR10,210 million).

2. Based on SAP's balance sheets presented earlier, which of the following liquidity ratios decreased in 2017 relative to 2016? (Note: More than one answer may be correct.)

   A. Cash
   B. Quick
   C. Current

   Hide Solution

   Solution:

   A, B, and C are correct. The cash ratio, quick ratio, and current ratio are lower in 2017 than in 2016.

Liquidity Ratios	Calculation	2017 EUR in millions	2016 EUR in millions
Current	Current assets ÷ Current liabilities	EUR11,930 ÷ EUR10,210 **= 1.17**	EUR11,564 ÷ EUR9,674 **= 1.20**
Quick (acid test)*	(Cash + Marketable securities +	(EUR4,011 + EUR990 + EUR5,899) ÷	(EUR3,702 + EUR1,124 + EUR5,924) ÷

Cash*	(Cash + Marketable securities) ÷ Current liabilities	(EUR4,011 + EUR990 ÷ EUR10,210 **= 0.49**	(EUR3,702 + EUR1,124 ÷ EUR9,674 **= 0.50**

* Marketable securities is assumed to be equal to Other Financial Assets as shown in Exhibit 6.

3. Based on SAP's balance sheets presented earlier, which of the following leverage ratios decreased in 2017 relative to 2016? (Note: more than one answer may be correct.)

A. Debt-to-equity.
B. Financial leverage.
C. Long-term debt-to-equity.

Hide Solution

Solution:

A, B, and C are correct. All three leverage ratios decreased in 2017 relative to 2016.

Solvency Ratios

Long-term debt-to-equity	Total long-term debt ÷ Total equity	EUR5,034 ÷ EUR25,540 **= 19.7%**	EUR6,481 ÷ EUR26,397 **= 24.6%**
Debt-to-equity	Total debt ÷ Total equity	(EUR1,561 + EUR5,034) ÷ EUR25,540 **= 25.8%**	(EUR 1,813 + EUR6,481) ÷ EUR26,397 **= 31.4%**
Financial Leverage	Total assets ÷ Total equity	EUR42,497 ÷ EUR25,540 **= 1.66**	EUR44,277 ÷ EUR26,397 **= 1.68**

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Cross-sectional financial ratio analysis can be limited by differences in accounting methods. In addition, lack of homogeneity of a company's operating activities can limit comparability. For diversified companies operating in different industries, using industry-specific ratios for different lines of business can provide better comparisons. Companies disclose information on operating segments. The financial position and performance of the operating segments can be compared to the relevant industry.

Ratio analysis requires a significant amount of judgment. One key area requiring judgment is understanding the limitations of any ratio. The current ratio, for example, is only a rough measure of liquidity at a specific point in time. The ratio captures only the amount of current assets, but the components of current assets differ significantly in their nearness to cash (e.g., marketable securities versus inventory). Another limitation of the current ratio is its sensitivity to end-of-period financing and operating decisions that potentially can affect current asset and current liability amounts. Another overall area requiring judgment is determining whether a ratio for a company is within a reasonable range for an industry. Yet another area requiring judgment is evaluating whether a ratio signifies a persistent condition or reflects only a temporary condition. Overall, evaluating specific ratios requires an examination of the entire operations of a company, its competitors, and the external economic and industry setting in which it is operating.

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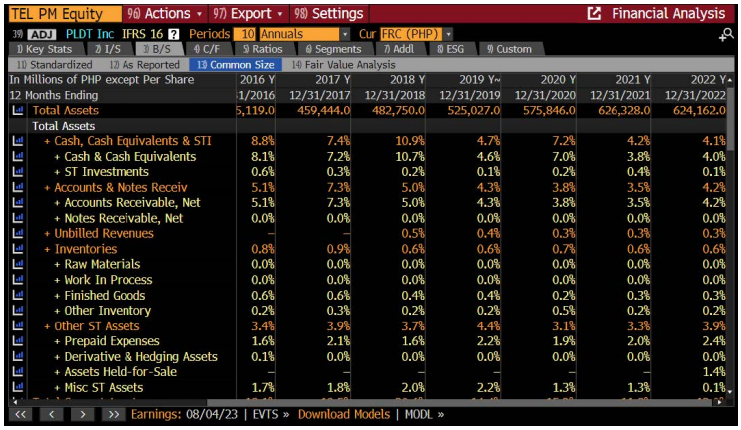

In Millions of PHP except Per Share	2016 Y	2017 Y	2018 Y	2019 Y	2020 Y	2021 Y	2022 Y
12 Months Ending	1/2016	12/31/2017	12/31/2018	12/31/2019	12/31/2020	12/31/2021	12/31/2022
Total Assets	5,119.0	459,444.0	482,750.0	525,027.0	575,846.0	626,328.0	624,162.0
Total Assets							
+ Cash, Cash Equivalents & STI	8.8%	7.4%	10.9%	4.7%	7.2%	4.2%	4.1%
+ Cash & Cash Equivalents	8.1%	7.2%	10.7%	4.6%	7.0%	3.8%	4.0%
+ ST Investments	0.6%	0.3%	0.2%	0.1%	0.2%	0.4%	0.1%
+ Accounts & Notes Receiv	5.1%	7.3%	5.0%	4.3%	3.8%	3.5%	4.2%
+ Accounts Receivable, Net	5.1%	7.3%	5.0%	4.3%	3.8%	3.5%	4.2%
+ Notes Receivable, Net	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
+ Unbilled Revenues	–	–	0.5%	0.4%	0.3%	0.3%	0.3%
+ Inventories	0.8%	0.9%	0.6%	0.6%	0.7%	0.6%	0.6%
+ Raw Materials	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
+ Work In Process	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
+ Finished Goods	0.6%	0.6%	0.4%	0.4%	0.2%	0.3%	0.3%
+ Other Inventory	0.2%	0.3%	0.2%	0.2%	0.5%	0.2%	0.2%
+ Other ST Assets	3.4%	3.9%	3.7%	4.4%	3.1%	3.3%	3.9%
+ Prepaid Expenses	1.6%	2.1%	1.6%	2.2%	1.9%	2.0%	2.4%
+ Derivative & Hedging Assets	0.1%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
+ Assets Held-for-Sale	–	–	–	–	–	–	1.4%
+ Misc ST Assets	1.7%	1.8%	2.0%	2.2%	1.3%	1.3%	0.1%

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NN Why does the first box question consider Total Financial Liabilities as only 5034, but ratio analysis question 3 consider total financial debt as 5034+1561?

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